

Technische Universität Chemnitz

# Exploring Reinforcement learning strategies in automated wheel truing

Master's Thesis

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# Abstract

This thesis attempts to find a solution to automatic spoked wheel truing through reinforcement learning in a simulated environment. We test and evaluate a set of different algorithms, state-spaces and reward schemes.

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# Chapter 1

## Introduction

### 1.1 Motivation

short motivation preferably relating the problem to material stress optimization problems.

### 1.2 Wheel truing

describing the problem and mention sim-to-real outlook

### 1.3 Thesis Structure

# Chapter 2

## Theory

TODO:

- start algorithm chapters (bullet points)
- finish physics section

**OVERVIEW:** This chapter should outline the necessary theoretical background on deep reinforcement learning, as well as the relevant physics from Dr. Ford's thesis on bicycle wheel dynamics.

**OPEN QUESTIONS:**

### 2.1 Reinforcement learning basics

Here I want to cover the basic ideas, definitions and mathematical prerequisites. This will probably include:

- Markov states
- states, actions and their values
- value iteration and policies
- function approximation and deep learning
- data sampling and replay buffer

After these basics, I want to describe the main ideas and various improvements of the different approaches of the three selected algorithms in the next subsections:

## 2.2 DeepQ (Rainbow)

## 2.3 Policy gradients (PPO)

## 2.4 World-Models (TDMPC)

## 2.5 Wheel Truing Physics

The environment used in this thesis is built upon a simulation based on a physical model of spoked bicycle wheels presented in [Ford2018ReinventingTW]. This section covers a brief overview of the problem formulation and numerical solution to the spoked wheel system.

### 2.5.1 Structural components and modeling assumptions

The spoked wheel is essentially built from three different components: the hub, rim and a number of spokes. Relative to the spokes and rim, the hub is a rigid structure of high stiffness; it is therefore modeled as stationary and non-deformable and provides the boundary condition of the spoke-rim system through its diameter  $d_h$  and width  $w_h$ . The rim is in essence a thin-walled, curved beam with constant radius  $R$  measured at the shear center of the cross-section. This cross-section features a complex cross-sectional geometry exhibiting monosymmetric geometry (symmetric about the radial axis but not the lateral axis) which introduces coupling between lateral bending and torsional deformations. To describe these interactions, the rim is modeled as being made from a linearly elastic material obeying the generalized Hooke's law, which forms a beam obeying Euler-Bernoulli theory. This boils down to neglecting shear deformations that are not associated with uniform torsion (plane sections remain plane and perpendicular to the neutral axis during bending) and ignoring in-plane distortion of the rim cross-section. From Hooke's law we need the Young's and shear moduli  $E$  and  $G$  respectively. The spokes are modeled as ideal, linear elastic bar elements, deforming only along their length with a Young's modulus  $E_s$ . The spokes are connected to the rim and hub by idealized, moment-free, ball-and-socket joints, meaning they do not transmit bending moments or torques. To describe deformations in the wheel, a local cylindrical coordinate system is defined through a right-handed orthonormal triad. The basis vectors  $\{e_1, e_2, e_3\}$  of this triad point in lateral (out of plane), radial (toward hub) and tangential (increasing arc length) directions. The origin travels with the rim centroid specified by an angular coordinate  $\theta \in [-\pi, \pi]$ . With this system we can describe the spoke position through  $\theta$  and directions through direction cosines  $\{c_1, c_2, c_3\}$  defined by the spoke unit vector  $n = c_1 e_1 + c_2 e_2 + c_3 e_3$  with  $c_1^2 + c_2^2 + c_3^2 = 1$ .

### 2.5.2 Displacement Field

The displacement of the rim is described by its motion and rotation of the shear center. The displacement of the shear center is expressed in the local coordinate basis:

- $u(\theta)$ : Lateral displacement in  $e_1$  direction (parallel to hub axis)
- $v(\theta)$ : Radial displacement in  $e_2$  direction (positive inward)
- $w(\theta)$ : Tangential displacement in  $e_3$  direction (along rim circumference)

The rotation is characterized by a single angle:

- $\Phi(\theta)$ : Twist about  $e_3$  axis (normal to cross-section)

These components collect into an augmented displacement vector:

$$d = [u, v, w, \phi]^T \quad (2.1)$$

Which is the fundamental unknown of the governing equations. At any arbitrary point, the displacement field can be obtained by adding the rigid-body rotation and warping contributions:

$$u(\theta, x, y) = u_s + \omega \times r + \frac{\phi'}{R} \alpha_s e_3 \quad (2.2)$$

where  $u_s$  is the shear center displacement,  $\omega$  is the rotation vector,  $r = [x, y - y_0, 0]^T$  is the position vector from the shear center to the point of interest,  $\alpha_s$  is the warping function, and  $y_0$  is the offset between the centroid and shear center.

### 2.5.3 Strain Energy

The strain energy stored in the rim due to deformation provides the starting point for deriving the equations of motion. By Hamilton's principle or the principle of virtual work, the equilibrium configuration minimizes the total potential energy (strain energy minus work done by external forces). The infinitesimal rotation angles of the basis vectors, to first order, are:

$$\omega_1 = v' + \frac{\omega}{R}, \omega_2 = u', \omega_3 = \phi \quad (2.3)$$

where primes denote differentiation with respect to arc length. The curvature changes associated with bending and twisting are:

$$\kappa_1 = \frac{u'}{R} = \frac{1}{R} \frac{du}{d\theta}, \kappa_2 = -\frac{v'}{R} = -\frac{1}{R} \frac{dv}{d\theta}, \kappa_3 = \frac{\phi'}{R} = \frac{1}{R} \frac{d\phi}{d\theta} \quad (2.4)$$

The longitudinal strain at a point in the cross-section is given by:

$$\epsilon_{33} = \epsilon_c - x\kappa_a - y\kappa_2 + \alpha_s\kappa_3 \quad (2.5)$$

where  $\epsilon_c$  is the strain at the centroid. The first two terms represent the contribution from bending (with the familiar linear variation across the section), while the last term accounts for the effect of non-uniform torsion (warping). The shear strain from uniform torsion is:

$$\gamma = r\left(\phi' - \frac{u'}{R}\right) \quad (2.6)$$

where  $r$  is the distance from the shear center. This strain is responsible for the torsional stiffness of the rim. The strain energy is obtained by integrating the strain energy density  $\frac{1}{2}(E\epsilon_{33}^2 + G\gamma^2)$  over the rim volume:

$$U_{rim} = \frac{1}{2} \int_0^{2\pi R} [EA\epsilon_c^2 + EI_1\kappa_1^2 + EI_2\kappa_2^2 + GJ\kappa_3^2 + EI_\omega(\kappa_3')^2] ds \quad (2.7)$$

# Chapter 3

## Methods

### 3.1 Environment

**TODO:**

- introduce the thinking behind different reward functions
- motivate and list wheel parameters (done)
- make sure the wheel parameters have the right units and descriptions (sorta done needs last look)
- in reward section introduce terminating and penalty condition (done)

**OVERVIEW:** Here I want to describe the process and considerations taken to transform the wheel simulation into a suitable reinforcement learning environment. Loosely, this will include:

- Isolating and optimizing the necessary physics calculations
- State-space design ([n]rimpoints, spoke tensions and Fourier coefficients)
- Action-space formulation (continuous one-hot, discrete and all spokes)
- Reward function design
- Episode structure and termination conditions

**OPEN QUESTIONS:**

- should we add a reward function similar to raw as the sum over turns? or even
- a mixed reward function combining spoke turns and rim displacement?

### 3.1.1 On Wheel Parameters

From the previous chapter we can infer that we have quite a few open parameters describing the wheel:

Table 3.1: Wheel simulation parameters

| Parameter             | Value                | Description                                |
|-----------------------|----------------------|--------------------------------------------|
| n_spokes              | 36                   | Number of spokes                           |
| max_tension_threshold | 1200 N               | Maximum allowable spoke tension            |
| max_tension_penalty   | true                 | Enable penalty for exceeding max tension   |
| hub_width             | 50 mm                | Hub width (distance between flanges)       |
| hub_diameter          | 40 mm                | Hub flange diameter                        |
| rim_radius            | 300 mm               | Rim radius (at shear center)               |
| rim_area              | 100 mm <sup>2</sup>  | Rim cross-sectional area                   |
| rim_I_lat             | 1500 mm <sup>4</sup> | Rim lateral moment of inertia ( $I_2$ )    |
| rim_I_rad             | 3000 mm <sup>4</sup> | Rim radial moment of inertia ( $I_1$ )     |
| rim_J_tor             | 500 mm <sup>4</sup>  | Rim torsion constant ( $J$ )               |
| rim_young_mod         | 69 GPa               | Rim Young's modulus (aluminum)             |
| rim_shear_mod         | 26 GPa               | Rim shear modulus                          |
| rim_I_warp            | 0.0 mm <sup>6</sup>  | Rim warping constant ( $I_w$ )             |
| spokes_crossings      | 3                    | Number of spoke crossings (lacing pattern) |
| spokes_diameter       | 2.0 mm               | Spoke wire diameter                        |
| spokes_young_mod      | 210 GPa              | Spoke Young's modulus (steel)              |
| number_modes          | 40                   | Number of Fourier modes ( $N$ )            |
| init_tension          | 800 N                | Initial spoke pretension ( $T_p$ )         |

### 3.1.2 On the Starting State

On reset of the environment, i.e., start of a new episode, we need to construct an out-of-true wheel. We do this by starting with a true wheel with completely uniform tensions and then bringing it out of true by choosing  $n_{\text{choice}}$  spokes and turning them some random uniformly sampled value from  $[-\text{max}_{\text{turn}}, \text{max}_{\text{turn}}]$ . We choose somewhat arbitrarily  $n_{\text{choice}} = 5$  and  $\text{max}_{\text{turn}} = 1$ .

### 3.1.3 On State-Space Formulations

The complete state is defined by the displacement field and spoke tensions in equilibrium as explained in the previous chapter. For representing the state we can choose an arbitrary number of points around the rim in terms of the rotation angle  $\theta$ . We make this choice to be 360 points, equidistant around the circle. For each of these points, we can simulate the displacement triad, giving us the local lateral, tangential and radial displacement. As it is not realistic to discern the tangential displacement by eye when truing a wheel, or even measure it through sensors, it is omitted from the state. The

spoke tensions in equilibrium are calculated by the simulation in units of Newtons. To reduce the variation in absolute values in the state (displacement vectors on the order of a few millimeters vs. spoke tensions on the order of hundreds of Newtons), the tensions are normalized by dividing through the initial tension. Since the displacement around the wheel is periodic by nature, we can also express this displacement in terms of its Fourier coefficients. For the lateral and radial displacement fields  $u_{\text{lat}}(\theta)$  and  $u_{\text{rad}}(\theta)$ , we compute the discrete Fourier transform coefficients up to  $N$  harmonics:

$$c_k = \frac{1}{M} \sum_{j=0}^{M-1} u(\theta_j) e^{-2\pi i k j / M}, \quad k = 0, 1, \dots, N \quad (3.1)$$

where  $M = 360$  is the number of sample points,  $\theta_j = 2\pi j / M$  are the equidistant angular positions, and  $u(\theta_j)$  represents either the lateral or radial displacement at position  $j$ . The Fourier representation consists of the real and imaginary parts of these coefficients for both displacement components, yielding a more compact state representation of  $4(N + 1)$  features. We choose  $N = 30$  by the elbow-graph heuristic on the distance of the original displacement points to the reconstructed points from the Fourier coefficients with some margin of safety. To explore how much of the full state is required for the agent to learn and if we could get away without measuring the spokes in a potential physical implementation, we combine these parts into the following state-space options:

- only displacement points
- displacement points and normalized spoke tensions
- Fourier coefficients
- Fourier coefficients and normalized spoke tensions

### 3.1.4 On Action-Space Formulations

Since DQN variants only work with discrete and TDMPC2 only with continuous action spaces, we need at least two formulations. Starting with the discrete formulation, we let the agent choose from  $[2 \cdot n_{\text{spokes}}]$  actions corresponding to a 0.1 turn in  $+/-$  direction of the corresponding spoke. We can transform this directly into a 2D continuous function by one-hot encoding the spoke vector and letting the second vector control the adjustment size. Additionally, we can also formulate a continuous action by letting the agent adjust all the spokes at once with a vector of length  $n_{\text{spokes}}$ , giving us the following three action spaces  $\mathcal{A}$ :

- **Continuous spoke selection:** two-dimensional continuous space

$$\mathcal{A}_{\text{continuous}} = [0, n_{\text{spokes}} - 1] \times [-1, 1] \subset \mathbb{R}^2 \quad (3.2)$$

where  $n_s$  is the number of spokes. An action  $a = (s, \Delta t) \in \mathcal{A}_{\text{continuous}}$  specifies a continuous spoke index  $s$  and a tension adjustment  $\Delta t$ .

- **Discrete spoke selection:**

$$\mathcal{A}_{\text{discrete}} = \{0, 1, 2, \dots, 71\} \quad (3.3)$$

representing 72 distinct action choices, where each action corresponds to a specific spoke-tension adjustment pair.

- **All spokes adjustment:**  $n_{\text{spokes}}$ -dimensional continuous space

$$\mathcal{A}_{\text{all\_spokes}} = [-1, 1]^{n_{\text{spokes}}} \subset \mathbb{R}^{n_s} \quad (3.4)$$

where an action  $a = (\Delta t_1, \Delta t_2, \dots, \Delta t_{n_s}) \in \mathcal{A}_{\text{all\_spokes}}$  specifies simultaneous tension adjustments for all spokes.

### 3.1.5 On Reward Functions

The formulation of the reward is notoriously important for successful reinforcement learning. (citations?)

We can evaluate the current state of the wheel by either looking at the remaining spoke turns from its initial true state or by measuring the total displacement by summing over the displacement vectors weighted by  $d\theta$ :

$$\|s_t\| \approx \sqrt{\sum_{k=1}^{N-1} \frac{\Delta\theta_k}{2} [\|u(\theta_k)\|_2^2 + \|u(\theta_{k+1})\|_2^2]} \quad (3.5)$$

where  $\|u(\theta)\|_2^2 = \sum_{d=1}^D u_d^2(\theta)$  is the squared Euclidean norm of the displacement vector at angle  $\theta$ , and define the step-improvement  $I_{\text{step}}$  as  $\|s_{t-1}\| - \|s_t\|$ .

- **Raw reward:** Directly penalizes the remaining displacement in the rim

$$r_t^{\text{raw}} = -\|s_t\| \quad (3.6)$$

- **Normalized reward:** Provides binary feedback based on step-wise improvement

$$r_t^{\text{normalized}} = \begin{cases} 0 & \text{if } I_{\text{step}} > 0 \\ -1 & \text{if } I_{\text{step}} \leq 0 \end{cases} \quad (3.7)$$

- **Spokes reward:** Provides binary feedback based on spoke adjustment magnitude

$$r_t^{\text{spokes}} = \begin{cases} 0 & \text{if } |\Delta t_i^{(t)}| \leq |\Delta t_i^{(t-1)}| \text{ for all } i \\ -1 & \text{otherwise} \end{cases} \quad (3.8)$$

where  $\Delta t_i^{(t)}$  denotes the turns applied to spoke  $i$  at time step  $t$ .

Additionally, we introduce two stopping conditions to reward the agent for producing a true wheel or punish for exceeding a tension threshold. Some research into actual manual wheel truing shows that the "trueness" is often evaluated only by the rim displacement with less concern for the uniformity of spoke tensions. The state of a true wheel is often seen as dependent on the material quality of the wheel. In this context, we often see discussion around the concepts of "decent," "good," and "golden" wheels, where these generally have the following specifications:

- A decent wheel refers to a common quality wheel which is considered reasonably true with a maximum rim displacement of 1–2 mm and a relative tension deviation of 15–20%
- A good wheel refers to a good quality wheel with a maximum rim displacement of 0.5–1 mm and a relative tension deviation of 5–10%
- A golden wheel refers to an excellent quality wheel with a maximum rim displacement of less than 0.5 mm and a relative tension deviation of under 5%

Since in simulation we can assume no errors or inherent tension in the material of the rim, we can orient ourselves close to the golden standard. From this we introduce the success condition as:

$$|u_d(\theta_i)| < 0.3 \text{ mm} \quad \forall d \in \{1, \dots, D\}, \forall i \in \{1, \dots, N\} \quad (3.9)$$

$$700 \text{ N} \leq T_j \leq 900 \text{ N} \quad \forall j \in \{1, \dots, n_s\} \quad (3.10)$$

$$\max_j T_j - \min_j T_j \leq 0.1 \cdot \bar{T} \quad (3.11)$$

where  $u_d(\theta_i)$  represents the displacement in dimension  $d$  at discrete angle  $\theta_i$ ,  $T_j$  is the tension in spoke  $j$ , and  $\bar{T} = \frac{1}{n_s} \sum_{j=1}^{n_s} T_j$  is the mean spoke tension. When these conditions are met, the episode terminates with a bonus reward of  $r_t = 20$ .

For the penalty condition we consider the fact that most wheel manufacturers give a maximum tension of 120 kg force on a spoke. We consider the episode as failed when a single spoke exceeds this threshold and we punish the agent with a reward of  $r_t = -20$ . To avoid filling the replay buffer with exceptionally bad states at the beginning, we also

terminate the episode after exceeding this threshold. So our second stopping condition can be formulated as:

$$\text{if } \exists j \in \{1, \dots, n_s\} : T_j > T_{\max} \quad \text{then} \quad r_t \leftarrow r_t - 20, \text{truncated} = \text{True} \quad (3.12)$$

where  $T_{\max}$  is the maximum tension threshold.

## 3.2 Experimental Design

This chapter describes the experimental methodology used to analyze the influence of state representation, action formulation, reward design, and optimization strategy on reinforcement learning performance in the wheel truing task. The experiments are structured as a controlled ablation study, followed by targeted hyperparameter optimization and optional investigations into temporal state augmentation.

The primary goal of the experimental design is not to obtain optimal absolute performance, but to systematically compare modeling and algorithmic choices in a physics-based reinforcement learning environment under a fixed computational budget.

### 3.2.1 Ablation Study Design

The ablation study forms the core of the experimental analysis. Its purpose is to identify which components of the reinforcement learning formulation are most influential for learning a successful wheel truing policy. To this end, multiple design dimensions are varied systematically while all other aspects of the environment and training procedure are held constant.

#### Ablation dimensions

The ablation study systematically varies the following components of the reinforcement learning setup:

- **State-space representation:** Rim displacement representations (point-based and Fourier-based), spoke-tension-only representations, and combined representations including both geometric and tension information.
- **Action-space formulation:** Discrete spoke adjustments, continuous single-spoke actions, and continuous multi-spoke adjustments.
- **Reward function design:** Reward formulations based on absolute rim displacement, step-wise improvement, and penalties related to spoke adjustment effort.

- **Reinforcement learning algorithm:** Value-based (DQN), policy-gradient (PPO), and model-based (TD-MPC2) approaches.

These dimensions were chosen to reflect both practical considerations in wheel truing (e.g., observability and control granularity) and common design choices in reinforcement learning. All experiments in the ablation study employ feedforward neural network policies without explicit recurrence. This restriction is intentional, as it allows the influence of state representation and reward design to be analyzed without the confounding effects of learned temporal memory. Temporal augmentation strategies are investigated separately in a later experimental phase.

### Controlled comparison strategy

To isolate the influence of individual design choices, experiments are conducted under a controlled comparison protocol. Wherever possible, only a single ablation dimension is varied at a time, while all other aspects of the training setup are held fixed.

Specifically, episode length, termination conditions, wheel parameters, and training budgets are identical across all configurations. Each configuration is trained once under stochastic but fixed conditions, reflecting a single realization of the learning process. Due to computational constraints imposed by the available resources, stochastic variability across multiple training runs is not explicitly characterized. Instead, the experimental design prioritizes breadth of methodological comparison over statistical estimation of expected performance. Conclusions are therefore drawn based on relative performance trends and qualitative behavioral differences between configurations rather than precise quantitative comparisons.

## 3.2.2 Training Protocol

### Training setup

Each configuration is trained for a fixed budget of  $1 \times 10^6$  environment steps. Episodes have a maximum length of 41 steps but may terminate earlier due to success or failure conditions defined by the environment.

Policies interact with the environment under identical termination and reward conditions. Exploration mechanisms are algorithm-specific and follow standard implementations.

Due to the computational cost of the physics-based simulation, each configuration is trained once.

### Logging and checkpoints

During training, the following quantities are logged:

- Episodic return
- Rim displacement metrics
- Spoke tension statistics
- Episode termination reason (success or failure)

For each training run, only the final model parameters are retained. No intermediate checkpoints are stored. This choice reflects the focus on final policy performance rather than learning dynamics.

### 3.2.3 Hyperparameter Optimization Strategy

#### Motivation

Hyperparameter sensitivity can significantly influence reinforcement learning performance. However, exhaustive tuning across all ablation configurations is computationally infeasible due to the high cost of the physics-based simulation and the fixed training budget of  $1 \times 10^6$  steps per run.

Therefore, hyperparameter optimization is applied selectively to a subset of promising configurations identified in the ablation study. For each optimization trial, models are trained once and evaluated only at the end of training.

#### Selection of candidates

Candidate configurations are selected based on:

- Training stability
- Success rate in producing a true wheel
- Final rim displacement and tension metrics

#### Optimization procedure

Hyperparameter optimization is performed using Optuna, employing Bayesian optimization to explore the parameter space efficiently. The search space is restricted to commonly tuned parameters such as learning rate, network width, and entropy regularization.

Each trial is trained for the same fixed number of environment steps and evaluated using the same metrics as in the ablation study to ensure comparability.

### 3.2.4 Evaluation Methodology

#### Training vs evaluation

To distinguish learning dynamics from final policy performance, evaluation is performed using frozen policies. During evaluation:

- Exploration noise is disabled
- Policies are tested on unseen initial wheel disturbances

#### Performance metrics

Performance is assessed using physically meaningful metrics:

- Root-mean-square and maximum rim displacement
- Maximum and relative spoke tension deviation
- Number of spoke adjustments required

These metrics reflect geometric accuracy, structural safety, and truing efficiency.

#### Aggregation and reporting

Results are reported per configuration without aggregation across multiple training seeds. As a consequence, conclusions are drawn based on relative trends, qualitative differences, and consistency across evaluation episodes rather than statistically significant performance differences.

Evaluation is performed across multiple unseen initial wheel disturbances, which provides insight into policy robustness despite the use of single training runs.

### 3.2.5 Temporal State Augmentation

Following the ablation study, selected configurations are further evaluated using temporal state augmentation techniques to address partial observability.

#### Frame stacking with action history

State vectors are augmented by stacking observations from previous time steps and including the previously applied action. This provides a short-term history without introducing recurrent architectures.

### Recurrent neural networks

For comparison, recurrent policies using gated recurrent units (GRUs) are evaluated on the same configurations. This allows a direct comparison between explicit temporal memory and recurrent state representations.

### 3.2.6 Methodological Limitations

The primary limitations of the methodology include:

- Use of a single training run per configuration
- Uncontrolled stochasticity during training
- Quasi-static simulation assumptions
- Absence of sim-to-real validation

As a result, conclusions focus on methodological trends and comparative behavior rather than statistically significant performance estimates.

# Chapter 4

## Results and Analysis

### 4.1 Baseline Performance

- performance of the algorithms in phase 1
- what can we say about the influence of reward functions and
- state space
- action space
- cross factor analysis

### 4.2 hyper-parameter optimization

I am not sure yet which candidates will do well and what exactly I want to evaluate here after optimizing the best candidates.

## Chapter 5

### Discussion and Conclusion

# Appendix

## **.1 Additional Figures**

Additional figures that didn't fit in the main chapters.

## **.2 Additional Tables**

Additional tables with detailed data.

## **.3 Code Listings**

Extended code listings that are too long for the main text.

## **.4 Experimental Details**

Detailed experimental configurations and parameters.